Antonio J. Ventimiglia

18162 Vestry Circle, Huntington Beach, California 92648
(714) 642-6491 ventimaj@icloud.com
LinkedIn: http://www.linkedin.com/in/antonio-j-ventimiglia Portfolio: https://tinyurl.com/AntonioVentimigliaPortfolio

EDUCATION

California Polytechnic State University, San Luis Obispo September 2022 - June 2026

- B.S. Mechanical Engineering, Concentration in Mechatronics
- Minor in Computer Science
- Total GPA: 3.93
- President's List (x3)**:** Dean's Honors List for Fall, Winter, and Spring quarters consecutively
- Mechatronics, System Dynamics, Fluid Dynamics, Heat Transfer / Thermal System Design, Mechanical Vibrations, Design for Strength and Stiffness, Artificial Intelligence, Deep Learning, Manufacturing: Joining/Subtractive Processes

TECHNICAL EXPERIENCE

General Dynamics Mission Systems: June 2024 - August 2025

Mechanical Engineer Intern: Space Avionics & Payloads

- Performed steady-state thermal analyses in Ansys Mechanical to support proposal-stage project qualification
- Designed a large-scale, 100+ component proof-of-concept assembly in Creo; pitched to the department's mechanical lead
- Completed mechanical qualification for over 650 high-reliability electrical components for flight PCBs
- Worked with several customer projects providing venting, thread engagement, and mass estimation analyses
- Collaborated with manufacturing to design versatile testing fixtures to remedy thermal conductivity and modularity issues
- Managed documentation configuration and control, ensuring accurate versioning and compliance with project specs

SLO Propulsion Technologies (SPT): September 2023 - Present

Project Lead, Throttleable Pintle Injector March 2025 - June 2025

- Designed and prototyped a linearly actuated pintle injector designed for 3:1 thrust throttleability
- Cost-efficient design requiring two-way communication between stepper motor and Arduino microcontroller

System Lead, Thrust Chamber Assembly September 2024 - Present

- Designed, prototyped, and manufactured chamber assembly for bi-propellant rocket. Target altitude of 38,000ft
- Research and selected ablative material, weighing mechanical properties against estimated price and lead times

Member, Fuel Injector September 2023 - September 2024

- Designed, prototyped, and manufactured a coaxial swirl injector for a 500 lbf thruster engine
- Utilized ANSYS modeling software to perform CFD: used learned flow characteristics to improve propellant impingement

Robotics: Autonomous Projectile Interception QArm January 2026 - Present

- Utilizing RealSense RGBD camera to track and locate a ball in space w.r.t to the robot arm to estimate trajectory (Kalman)
- DH parameterization of 4-DOF arm to perform forward/reverse kinematics. Singularity and system workspace defining.
- Joint space manipulation with fifth-order spline trajectories. Sensing uncertainty and end effect pose padding for stability
- OOP software architecture with multithreading. Enabled complex image processing and constant robot command/updating

Mechatronics: Romi - Obstacle Course Traverser September 2025 - December 2025

- Built autonomous differential-drive Romi robot to traverse a waypoint-based obstacle course using IR line-sensor array
- Developed a priority-based, round-robin multitasking firmware architecture with inter-task state-machine coordination
- Implemented closed-loop motor speed and heading control and a discrete state-space observer to estimate global pose
- Integrated Bluetooth command + data streaming via UART and IMU orientation via I2C, enabling runtime calib and tuning

Senior Design: Avian Detection Module: September 2025 - Present

- Designing, manufacturing, and testing a sensor module that detects the locations of avian targets in various environments
- Led hardware integration and software development. Defining comm. protocols between peripherals and external devices
- Utilizing deep learning models (CNNs) to classify and locate avian targets in frame. Stereo camera for depth perception
- Implementing a scheduler-based software architecture to balance peripheral device task execution and timing

SKILLS AND QUALITIES

ANSYS Mechanical, Creo, NX, GD&T, FEA, EES, PDM, Windchill, SOLIDWORKS, Fusion 360, Python, PyTorch, Java, MATLAB, Simulink, Assembly (RISC), C++, ROS, SPI, I2C, UART, Microsoft Office, Mill (manual + CNC), Lathe, TIG/MIG welding, Spanish (Working Proficiency), French (Limited Proficiency), Leadership Skills, Decisive, Project Management

PERSONAL INTERESTS

Baseball Player: Played competitive baseball for 15 years, traveled across the country to play in tournaments 2007 - 2022
Investing: Started trading at 15 years old 2019 - Present
Quantum Mechanics: A lover and avid student of topics like QFT, (Special) Relativity, & quantum chromodynamics 2018 - Present
Poker: Watched poker on TV while growing up, began playing with friends in high school 2020 - Present

GENERAL DYNAMICS
Mission Systems

SPT